CS 767 Advanced Machine Learning Homework 3

Joseph Rissman

U09545917

1. **Pandas practice.** See .ipynb file and related .pdf file.
2. **Seaborn.** See .ipynb file and related .pdf file.
3. **(Matrix multiplication). Let**
   $A = [6, -2, -2; 10, -3, 1; -10, 5, 1]_{3\times3}$
   $B = [9, 4, -4; 4, 7, 0; -4, 0, 11]_{3\times3}$
   $C = [3, 1; 0, -2; 4, 0]_{3\times2}$
   $\boldsymbol{a} = [5; 1; 2\,]_{3\times1}$
   $\boldsymbol{b} = [3, 0, 8]_{1\times3.}$
   Here, rows of the matrix are separated by semicolons, so, for example, $[6, -2, -2]$ is the first row of A. Pay very close attention to the **matrix sizes**.
   Calculate the following or explain why they cannot be calculated.
   a. $A\boldsymbol{a}$, $A\boldsymbol{b}$, $A\boldsymbol{b}^T$, AB.

$$A\boldsymbol{a} = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6\times5-2\times1-2\times2 \\ 10\times5-3\times1+1\times2 \\ -10\times5+5\times1+1\times2 \end{bmatrix} = \begin{bmatrix} 24 \\ 49 \\ -43 \end{bmatrix}$$

   $A\boldsymbol{b}$ cannot be calculated because the matrices have incompatible dimensions.

$$A\boldsymbol{b}^T = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 6\times3-2\times0-2\times8 \\ 10\times3-3\times0+1\times8 \\ -10\times3+5\times0+1\times8 \end{bmatrix} = \begin{bmatrix} 2 \\ 38 \\ -22 \end{bmatrix}$$

$$AB = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 6\times9-2\times4-2\times-4 & 6\times4-2\times7-2\times0 & 6\times-4-2\times0-2\times11 \\ 10\times9-3\times4+1\times-4 & 10\times4-3\times7+1\times0 & 10\times-4-3\times0+1\times11 \\ -10\times9+5\times4+1\times-4 & -10\times4+5\times7+1\times0 & -10\times-4+5\times0+1\times11 \end{bmatrix}$$

$$= \begin{bmatrix} 54 & 10 & -46 \\ 74 & 19 & -29 \\ -74 & -5 & 51 \end{bmatrix}$$

   b. Compute $A\boldsymbol{a}$ in two different ways: matrix multiplication and weighted sum of columns.
   Matrix Multiplication:

$$A\boldsymbol{a} = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6\times5-2\times1-2\times2 \\ 10\times5-3\times1+1\times2 \\ -10\times5+5\times1+1\times2 \end{bmatrix} = \begin{bmatrix} 24 \\ 49 \\ -43 \end{bmatrix}$$

   Weighted Sum of Columns:

$$A\boldsymbol{a} = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = 5\begin{bmatrix} 6 \\ 10 \\ -10 \end{bmatrix} + 1\begin{bmatrix} -2 \\ -3 \\ 5 \end{bmatrix} + 2\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 30 \\ 50 \\ -50 \end{bmatrix} + \begin{bmatrix} -2 \\ -3 \\ 5 \end{bmatrix} + \begin{bmatrix} -4 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 24 \\ 49 \\ -43 \end{bmatrix}$$

   c. AB, BA, AA^T, A^TA

$$AB = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}\begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 6\times9-2\times4-2\times-4 & 6\times4-2\times7-2\times0 & 6\times-4-2\times0-2\times11 \\ 10\times9-3\times4+1\times-4 & 10\times4-3\times7+1\times0 & 10\times-4-3\times0+1\times11 \\ -10\times9+5\times4+1\times-4 & -10\times4+5\times7+1\times0 & -10\times-4+5\times0+1\times11 \end{bmatrix}$$

$$= \begin{bmatrix} 54 & 10 & -46 \\ 74 & 19 & -29 \\ -74 & -5 & 51 \end{bmatrix}$$

$$BA = \begin{bmatrix} 9 & 4 & -4 \\ 4 & 7 & 0 \\ -4 & 0 & 11 \end{bmatrix} \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 9 \times 6 + 4 \times 10 - 4 \times -10 & 9 \times -2 + 4 \times -3 - 4 \times 5 & 9 \times -2 + 4 \times 1 - 4 \times 1 \\ 6 \times 4 + 10 \times 7 - 10 \times 0 & -2 \times 4 - 3 \times 7 + 5 \times 0 & -2 \times 4 + 1 \times 7 + 1 \times 0 \\ 6 \times -4 + 10 \times 0 - 10 \times 11 & -2 \times -4 - 3 \times 0 + 5 \times 11 & -2 \times -4 + 1 \times 0 + 1 \times 11 \end{bmatrix}$$
$$= \begin{bmatrix} 134 & -50 & -18 \\ 94 & -29 & -1 \\ -134 & 63 & 19 \end{bmatrix}$$
$$AA^T = \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix} \begin{bmatrix} 6 & 10 & -10 \\ -2 & -3 & 5 \\ -2 & 1 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 6 \times 6 - 2 \times -2 - 2 \times -2 & 6 \times 10 - 2 \times -3 - 2 \times 1 & 6 \times -10 - 2 \times 5 - 2 \times 1 \\ 10 \times 6 - 3 \times -2 + 1 \times -2 & 10 \times 10 - 3 \times -3 + 1 \times 1 & 10 \times -10 - 3 \times 5 + 1 \times 1 \\ -10 \times 6 + 5 \times -2 + 1 \times -2 & -10 \times 10 + 5 \times -3 + 1 \times 1 & -10 \times -10 + 5 \times 5 + 1 \times 1 \end{bmatrix}$$
$$= \begin{bmatrix} 44 & 64 & -72 \\ 64 & 110 & -114 \\ -72 & -114 & 126 \end{bmatrix}$$
$$A^T A = \begin{bmatrix} 6 & 10 & -10 \\ -2 & -3 & 5 \\ -2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & -2 \\ 10 & -3 & 1 \\ -10 & 5 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} 6 \times 6 + 10 \times 10 - 10 \times -10 & 6 \times -2 + 10 \times -3 - 10 \times 5 & 6 \times -2 + 10 \times 1 - 10 \times 1 \\ -2 \times 6 - 3 \times 10 + 5 \times -10 & -2 \times -2 - 3 \times -3 + 5 \times 5 & -2 \times -2 - 3 \times 1 + 5 \times 1 \\ -2 \times 6 + 1 \times 10 + 1 \times -10 & -2 \times -2 + 1 \times -3 + 1 \times 5 & -2 \times -2 + 1 \times 1 + 1 \times 1 \end{bmatrix}$$
$$= \begin{bmatrix} 236 & -92 & -12 \\ -92 & 38 & 6 \\ -12 & 6 & 6 \end{bmatrix}$$

d. Can you say something about the symmetry properties of matrices like $A^T A$ and AA^T?
The matrices $A^T A$ and AA^T are symmetric along the diagonal, meaning the transpose of each matrix is equal to the original matrix. This can be proven using matrix algebra:
$(A^T A)^T = (A^T)(A^T)^T$ using the property $(AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}}$,
$(A^T)(A^T)^T = A^T A$ using the property $(A^T)^T = A$, therefore
$(A^T A)^T = A^T A$.
The same proof works for AA^T:
$AA^T = (A^T)^T A$ using the property $(AB)^{\mathrm{T}} = B^{\mathrm{T}} A^{\mathrm{T}}$,
$(A^T)^T (A^T) = AA^T$ using the property $(A^T)^T = A$, therefore
$(AA^T)^T = AA^T$.
This means that the matrices $A^T A$ and AA^T will always be symmetric matrices.

e. $\boldsymbol{ab}$, $\boldsymbol{ba}$, $(\boldsymbol{ab})A$, $\boldsymbol{a}(\boldsymbol{b}A)$
$$\boldsymbol{ab} = \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 5 \times 3 & 5 \times 0 & 5 \times 8 \\ 1 \times 3 & 1 \times 0 & 1 \times 8 \\ 2 \times 3 & 2 \times 0 & 2 \times 8 \end{bmatrix} = \begin{bmatrix} 15 & 0 & 40 \\ 3 & 0 & 8 \\ 6 & 0 & 16 \end{bmatrix}$$
$$\boldsymbol{ba} = \begin{bmatrix} 3 & 0 & 8 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} = [3 \times 5 + 0 \times 1 + 8 \times 2] = [31]$$

f. Compute $\|\boldsymbol{a} - \boldsymbol{b}^T\|_2^2$ How will you express this in terms of $\|\boldsymbol{a}\|_2^2$, $\|\boldsymbol{b}\|_2^2$, and $\langle \boldsymbol{a}, \boldsymbol{b} \rangle$?
$$\|\boldsymbol{a} - \boldsymbol{b}^T\|_2^2 = \left\| \begin{bmatrix} 5 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix} \right\|_2^2 = 2^2 + 1^2 + (-6)^2 = 41$$
$$\|\boldsymbol{a} - \boldsymbol{b}^T\|_2^2 = \|\boldsymbol{a}\|_2^2 + \|\boldsymbol{b}\|_2^2 - 2\langle \boldsymbol{a}, \boldsymbol{b} \rangle$$

4. Write elements of the first column of the product AB (given in Q3) in terms of row and column vectors. Recall that the inner product between two column vectors x, y can be written as $x^T y$, i.e., $\langle x, y \rangle = x^T y$.

$$AB_{1,1} = \langle \begin{bmatrix} 6 \\ -2 \\ -2 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -4 \end{bmatrix} \rangle \qquad AB_{2,1} = \langle \begin{bmatrix} 10 \\ -3 \\ 5 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -4 \end{bmatrix} \rangle \qquad AB_{3,1} = \langle \begin{bmatrix} -10 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ -4 \end{bmatrix} \rangle$$

5. **(Triangular matrices).** Suppose U_1, U_2 are two upper triangular matrices and L_1, L_2 are two lower triangular matrices. Using examples (3× 3 matrices will suffice), explain which of the following are triangular: $U_1 + U_2, U_1U_2, U_1 + L_1, L_2^2, L_1L_2$.

Suppose that:

$$U_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, U_2 = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix}, L_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, L_2 = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

$U_1 + U_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$ the result is an upper triangular matrix.

$U_1U_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1\times 2+1\times 0+1\times 0 & 1\times 2+1\times 2+1\times 0 & 1\times 2+1\times 2+1\times 2 \\ 0\times 2+1\times 0+1\times 0 & 0\times 2+1\times 2+1\times 0 & 0\times 2+1\times 2+1\times 2 \\ 0\times 2+0\times 0+1\times 0 & 0\times 2+0\times 2+1\times 0 & 0\times 2+0\times 2+1\times 2 \end{bmatrix} =$ $\begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$ the result is an upper triangular matrix.

$U_1 + L_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ the result is not a triangular matrix.

$L_2^2 = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 4+0+0 & 0+0+0 & 0+0+0 \\ 4+4+0 & 0+4+0 & 0+0+0 \\ 4+4+4 & 0+4+4 & 0+0+4 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 8 & 4 & 0 \\ 12 & 8 & 4 \end{bmatrix}$ the result is a lower triangular matrix.

$L_1L_2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 2+0+0 & 0+0+0 & 0+0+0 \\ 2+2+0 & 0+2+0 & 0+0+0 \\ 2+2+2 & 0+2+2 & 0+0+2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 0 \\ 6 & 4 & 2 \end{bmatrix}$ the result is a lower triangular matrix.

6. Given a vector $\boldsymbol{u} = [u_1, u_2, \dots, u_n]$, what can you say about the norm of the vector

$$\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}$$

Explain how you arrived at your answer in detail.

The norm of the vector $\hat{\boldsymbol{u}}$, which is denoted by $\|\hat{\boldsymbol{u}}\|$**,** will always be equal to one. This means that $\hat{\boldsymbol{u}}$ is a unit vector. This can be proven by starting with the definition of the vector

$$\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}$$

Take the norm of both sides of the equation

$$\|\hat{\boldsymbol{u}}\| = \left\|\frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}\right\|$$

Now use the property of norms with scalars, where for any scalar c and vector v, the norm $\|cv\| = |c| \times \|v\|$. In the above equation, $\frac{1}{\|\boldsymbol{u}\|}$ is a positive scalar (from the definition of norms). Therefore

$$\|\hat{\boldsymbol{u}}\| = \left(\frac{\mathbf{1}}{\|\boldsymbol{u}\|}\right)\|\boldsymbol{u}\|$$

The norm $\|\boldsymbol{u}\|$ in the numerator and denominator cancel, leaving $\|\hat{\boldsymbol{u}}\| = \mathbf{1}$**.**

In simpler terms, $\hat{\boldsymbol{u}}$ is a unit vector because it is defined to be a vector divided by that vector's magnitude. Any vector that is divided by its magnitude will be scaled to have a magnitude of one, making it a unit vector.

An alternative proof can be constructed by examining the elements of the vector $\hat{\boldsymbol{u}}$.

$$\boldsymbol{u} = [u_1, u_2, \dots, u_n]$$

$$\|\boldsymbol{u}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

$$\|\boldsymbol{u}\|^2 = u_1^2 + u_2^2 + \cdots + u_n^2$$

$$\hat{\boldsymbol{u}} = \left[\frac{u_1}{\|\boldsymbol{u}\|}, \frac{u_2}{\|\boldsymbol{u}\|}, \dots, \frac{u_n}{\|\boldsymbol{u}\|}\right]$$

$$\|\widehat{\boldsymbol{u}}\| = \sqrt{\left(\frac{u_1}{\|\boldsymbol{u}\|}\right)^2 + \left(\frac{u_2}{\|\boldsymbol{u}\|}\right)^2 + \cdots + \left(\frac{u_n}{\|\boldsymbol{u}\|}\right)^2}$$

$$\|\widehat{\boldsymbol{u}}\| = \sqrt{\frac{u_1^2}{\|\boldsymbol{u}\|^2} + \frac{u_2^2}{\|\boldsymbol{u}\|^2} + \cdots + \frac{u_n^2}{\|\boldsymbol{u}\|^2}}$$

$$\|\widehat{\boldsymbol{u}}\| = \sqrt{\frac{u_1^2}{u_1^2 + u_2^2 + \cdots + u_n^2} + \frac{u_2^2}{u_1^2 + u_2^2 + \cdots + u_n^2} + \cdots + \frac{u_n^2}{u_1^2 + u_2^2 + \cdots + u_n^2}}$$

$$\|\widehat{\boldsymbol{u}}\| = \sqrt{\frac{u_1^2 + u_2^2 + \cdots + u_n^2}{u_1^2 + u_2^2 + \cdots + u_n^2}} = \sqrt{1}$$

$$\|\widehat{\boldsymbol{u}}\| = 1$$

7. What are antisymmetric matrices (note that they are also known as "skew-symmetric" matrices)? What can you say about the diagonal entries of an antisymmetric matrix?
   Antisymmetric matrices are matrices that satisfy the equation $A^T = -A$. Because the diagonal entries end up in the same position after transposition, they must be equal to their own negative. That means they must satisfy the equation $a_{i,i} = -a_{i,i}$. Since the only number that can satisfy that equation is zero, all diagonal entries of an antisymmetric matrix must be zero.
8. **(Orthogonality).** Find the L^2-norm of the following vectors $\boldsymbol{a} = [4,2,-6], \boldsymbol{b} = [16,-32,0]$. Are $\boldsymbol{a}, \boldsymbol{b}$ orthogonal vectors?
   $\|\boldsymbol{a}\|_2 = \sqrt{4^2 + 2^2 + (-6)^2} = \sqrt{16 + 4 + 36} = \sqrt{56} \approx 7.48$
   $\|\boldsymbol{b}\|_2 = \sqrt{16^2 + (-32)^2 + 0^2} = \sqrt{256 + 1024 + 0} = \sqrt{1280} \approx 35.77$
   $\langle \boldsymbol{a}, \boldsymbol{b} \rangle = 4 \times 16 + 2 \times -32 - 6 \times 0 = 64 - 64 + 0 = 0$
   The vectors are orthogonal because their inner product is zero.
9. **(Unit Vectors).** What is a unit vector? Find a unit vector orthogonal to $\boldsymbol{c} = [4, -3]$.
   A unit vector is any vector with a length of one. Let $\hat{u} = [u_1, u_2]$ be the unit vector orthogonal to $\boldsymbol{c}$**.** From the orthogonality rule, we know that $\langle \hat{u}, c \rangle = 0$. Therefore, $4u_1 - 3u_2 = 0$ gives the first equation. From the unit vector definition, we have $\|\hat{u}\|_2 = 1$. Therefore, $\sqrt{u_1^2 + u_2^2} = 1$ gives the second equation. Solve the second equation for u_2:

$$\sqrt{u_1^2 + u_2^2} = 1$$
$$u_1^2 + u_2^2 = 1^2$$
$$u_2^2 = 1 - u_1^2$$
$$u_2 = \sqrt{1 - u_1^2}$$

   Substitute u_2 into the first equation to get:

$$4u_1 - 3\sqrt{1 - u_1^2} = 0$$
$$4u_1 = 3\sqrt{1 - u_1^2}$$
$$\frac{4}{3}u_1 = \sqrt{1 - u_1^2}$$
$$\left(\frac{4}{3}u_1\right)^2 = 1 - u_1^2$$
$$\frac{16}{9}u_1^2 = 1 - u_1^2$$
$$\frac{25}{9}u_1^2 = 1$$
$$u_1^2 = \frac{9}{25}$$

$$u_1 = \frac{3}{5}$$

Substitute u_1 into the first equation to get:

$$4u_1 - 3u_2 = 0$$

$$4\left(\frac{3}{5}\right) - 3u_2 = 0$$

$$4 - 5u_2 = 0$$

$$5u_2 = 4$$

$$u_2 = \frac{4}{5}$$

Finally, the unit vector orthogonal to $\boldsymbol{c} = [4, -3]$ is $\hat{u} = \left[\frac{3}{5}, \frac{4}{5}\right]$.

10. **(Angle between vectors).** Let $a = [1,1,1], b = [2,3,1], c = [-1, 1, 0]$. Find the cosine of the angle between the vectors $\boldsymbol{a} + \boldsymbol{b}, \boldsymbol{c}$.
    Let the vector $\boldsymbol{d} = \boldsymbol{a} + \boldsymbol{b}$. The cosine of the angle between the vectors $\boldsymbol{c}, \boldsymbol{d}$ is given by the equation

$$\cos\theta = \frac{\boldsymbol{c} \cdot \boldsymbol{d}}{||\boldsymbol{c}|| \times ||\boldsymbol{d}||}$$

$$\boldsymbol{d} = \boldsymbol{a} + \boldsymbol{b} = [1,1,1] + [2,3,1] = [3,4,2]$$

$$\boldsymbol{c} \cdot \boldsymbol{d} = -1 \times 3 + 1 \times 4 + 0 \times 2 = -3 + 4 + 0 = 1$$

$$||\boldsymbol{c}|| = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{1 + 1 + 0} = \sqrt{2}$$

$$||\boldsymbol{d}|| = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

$$\cos\theta = \frac{\mathbf{1}}{\sqrt{2} \times \sqrt{29}} \approx 0.1313$$

11. (Projections). Find the vector closest to $\boldsymbol{a}$ on the axis through $\boldsymbol{b}$.
    a. $a = [1, 1, 3], b = [0, 0, 5]$

       $proj_b a = \left(\frac{b \cdot a}{||b||^2}\right) b$

       $b \cdot a = 0 \times 1 + 0 \times 1 + 5 \times 3 = 15$

       $||b||^2 = 0^2 + 0^2 + 5^2 = 25$

       $\left(\frac{b \cdot a}{||b||^2}\right) = \frac{15}{25} = \frac{3}{5}$

       $\left(\frac{b \cdot a}{||b||^2}\right) b = \frac{3}{5}[0 \quad 0 \quad 5] = [0 \quad 0 \quad 3]$

       $proj_b a = [0 \quad 0 \quad 3]$
    b. $a = [0, 4, -3], b = [0, 4, 3]$

       $proj_b a = \left(\frac{b \cdot a}{||b||^2}\right) b$

       $b \cdot a = 0 \times 0 + 4 \times 4 + 3 \times -3 = 7$

       $||b||^2 = 0^2 + 4^2 + 3^2 = 25$

       $\left(\frac{b \cdot a}{||b||^2}\right) = \frac{7}{25}$

       $\left(\frac{b \cdot a}{||b||^2}\right) b = \frac{7}{25}[0 \quad 4 \quad 3] = \left[0 \quad \frac{28}{25} \quad \frac{21}{25}\right] = [0 \quad 1.12 \quad 0.84]$

       $proj_b a = [0 \quad 1.12 \quad 0.84]$